

Chemical Engineering Curriculum

Curriculum Day May 17, 2022



BS Chemical Engineering Curriculum

Summary of Requirements

Nonengineering and General Education Requirements	70
Required in the College of Engineering	49
(Selective)	3
Technical Elective ^a	3
Electives outside the Major Rubric ^a	3
Total Hours	128



- ENGL160/161
- 5 General Education courses
- MATH 180,181,210,220
- PHYS 141, 142
- Non-ChE Elective
- CHEM 122/123, 124/125, 222, 232, 233, 234, 342, (346)

- ENGR100, ECE 210, CME 260, CS 109
- CHE Courses: 201, 205, 210, 230, 301, 311, 312, 313, 321, (330), 341, 381, 382, 396, 397, 499
- 400-level technical elective

Chemical Engineering Electives

- 392: Undergraduate Research
- 422: Biochemical Engineering
- 425: Nanotechnology for Pharmaceutical Applications
- 427: Entrepreneurship in Engineering
- 433: Process Simulation with Aspen Plus
- 438: Computational Molecular Modeling
- 440: Non-Newtonian Fluids
- 450: Air Pollution Engineering
- 451: Renewable Energy Technologies
- 453: Fundamentals of Electrochemistry
- 454: Molecular & Macromolecular Engineering
- 455: Nanoscale Systems in Chemical Engineering
- 457: Colloidal and Interfacial Phenomena
- 494: Clean Energy and the Electric Grid
- 494: Biopharmaceutical Process Engineering

Chemical Engineering Flowchart

<https://catalog.uic.edu/ucatalog/colleges-depts/engineering/che/bs-che/>

ENGR 100/101 Overview

- ENGR 100/101 is a coordinated effort by Engineering Undergraduate Affairs in conjunction with the six engineering departments at UIC: One faculty member from each department directs the department-specific sessions, with the day-to-day course topics facilitated by upper-level undergraduate student mentors.

Seminar Days and Times:

CRN 44483 Wednesday 9:00 - 9:50 AM

CRN 44493 Friday 12:00 - 12:50 PM

CRN 44545 Friday 4:00 - 4:50 PM

Week 1 Welcome and introductions. Discuss adding/dropping classes, navigating the degree audit/course plan. Assign course planning assignment, due following week

Week 2 “What is Chemical Engineering” video and presentation; discussion; **unit operations lab tour** Course planning assignment due.

Week 3 Professionalism presentation and discussion; TA video (Tips & Tricks); AIChE presentation

Week 4 Technical Project: Fermentation-Description and Assignment

Week 5 Project Work: Technical information, experimental setup

Week 6 Project Work: Perform experiments

Week 7 Project wrap-up; **Research Lab tours** Project report due

Week 8 Pre-Advising presentation, Department resources (Eduardo Martinez)

CHE 499 Overview

Specific goals for the course, Students will:

- Interact with practicing chemical engineers
- Learn about traditional and non-tradition chemical engineering career paths, both in academia and industry
- Learn job searching skills; resume writing, interviewing skills, salary negotiation
- Learn what industries hire chemical engineers
- Learn about their role as UIC Chemical Engineering alumni
- Relate their UIC educational experience in an exit survey
- Complete a written examination on engineering problems, careers, teamwork and ethics

Seminar Speakers: Fall 2021

September 10: Jonna McHugh, UIC Engineering Career Center

September 24: Robert Orent, Stepan

October 8: Thomas Scheller, ProLeiT

October 15: Rick Burns, Blake PincheckTetrapak

October 22: Chris Ciszek, CSL Behring

October 29: Lissa Aquino, Stepan Chemicals Safety/Environment November

5: Michael Bentel, Clemson University

November 12: Said Al-Hallaj, AllCellTechnology

November 19: Jeff Tyska, UOP

December 3: Jennifer Nacht, Cardinal Intellectual Property